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### Research Interests

**Embodied AI systems**, especially on vision-language-action (VLA) models, real-robot learning, first-person wearable agents, robot agents, and long-horizon manipulation.

**On-device AI systems**, especially on LLM inference acceleration (for phones), speech enhancement (for AI glasses), and tool-use agents (for personal computers).

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### Experiences

Nov 2024 – Present **Research Scientist in Central Software Institute,**  
*Huawei Technologies, Shenzhen, China*

- Embodied AI and Robot Agents
  - Lead a 6-person team to build [RobiAgent](#), a dual-arm robot agent prototype for studying real-robot manipulation and real-time, constraint-aware coordination of coupled robot subtasks.
  - Explore VLA-based robot manipulation through  $\pi_{0.5}$  [simulation](#) reproduction, [real-robot](#) data collection, imitation learning, and deployment on SO101 arms.
- Wearable Multimodal Intelligence on AI Glasses
  - Build [VisionBot](#), an AI-glasses-oriented agent for always-on perception and service, with demos including lost-and-found via continuous visual search and fitness coaching via first-person activity logging and feedback.
  - Develop a target speech hearing system for AI glasses by designing a stereo speech enhancement model, building a streaming inference pipeline, and deploying it through a HarmonyOS application.
- LLM/VLM Applications and Systems
  - Develop LLM/VLM applications including on-device fraud webpage detection, large-scale RAG-based question answering, and an MCP-compatible command-line agent ([Hey](#)).
  - Explore efficient inference for autoregressive language models via parallel decoding.

Sep 2019 – Jul 2024 **PhD Candidate in Computer Science and Engineering,**  
*The Hong Kong University of Science and Technology, Kowloon, Hong Kong*

- Private Inference for LLMs
  - Protect system prompt with minimized impact on conversational quality ([PromptKeeper](#))
- Private and Efficient Federated Training
  - Secure participant selection against adversarial servers ([Lotto](#))
  - Enable dropout-resilient differential privacy without loss in time performance ([Dordis](#))
  - Train asynchronously with principles for end-to-end speedup ([Pisces](#))

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### Education

Sep 2019–Jul 2024 **Ph.D. in [Computer Science and Engineering](#),**  
*The Hong Kong University of Science and Technology, Kowloon, Hong Kong*  
Dissertation: “[Towards Private and Efficient Cross-Device Federated Learning](#)”

Jul 2018 – Oct 2018 **Research Intern in [Prof. Dean Tullsen’s Research Group](#),**  
*University of California, San Diego, La Jolla, US*

- [Defense](#) against Return-Oriented Programming with Context-Sensitive Decoding on x86-64.

Sep 2015–Jun 2019 **B.Eng. in [Computer Science](#),**  
*Zhejiang University, Hangzhou, China*  
GPA: 3.97/4.0, Graduated with Outstanding Honor (Zhejiang Province)

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### Publications

#### Selected Publications

2025 **Zhifeng Jiang**, Zihua Jin, Guoliang He. “[PromptKeeper: Safeguarding System Prompts for LLMs](#)” , in the *Findings of the ACL EMNLP 2025*

- 2024 **Zhifeng Jiang**, Peng Ye, Shiqi He, Wei Wang, Ruichuan Chen, Bo Li. “[Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning](#)” , in the *Proc. of USENIX Security 2024*
- 2024 **Zhifeng Jiang**, Wei Wang, Ruichuan Chen. “[Dordis: Efficient Federated Learning with Dropout-Resilient Differential Privacy](#)” , in the *Proc. of ACM EuroSys 2024*
- 2023 **Zhifeng Jiang**, Wei Wang, Bo Li, Qiang Yang. “[Towards Efficient Synchronous Federated Training: A Survey on System Optimization Strategies](#)” , in *IEEE TBD, Volume 9, Issue 2*
- 2022 **Zhifeng Jiang**, Wei Wang, Baochun Li, Bo Li. “[Pisces: Efficient Federated Learning via Guided Asynchronous Training](#)” , in the *Proc. of ACM SoCC 2022*

#### Other Publications

- 2026 Na Lv, Jiayi Zhang, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Quan Chen, Minyi Guo. “[Enabling Client-Autonomous Training Optimizations for Efficient Federated Learning](#)” , accepted to appear in *IEEE TPDS*
- 2026 Zekai Su, Yongjun He, Jiqing Han, Jianchen Li, Yikun Jiang, **Zhifeng Jiang**, Tianhong Ding. “[AS-EVNorm: Tail-Aware Extreme Value Normalization for Speaker Verification](#)” , in *IEEE Signal Processing Letters, Volume 33*
- 2026 Jiayi Zhang, Zuo Gan, Chen Chen, **Zhifeng Jiang**, Hao Wang, Yifei Zhu, Quan Chen, Minyi Guo. “[Boosting Gradient-based Training Diagnosis for Efficient and Accurate Federated Learning](#)” , accepted to appear in *IEEE TMC*
- 2024 Peng Ye, **Zhifeng Jiang**, Wei Wang, Bo Li, Baochun Li. “[Feature Reconstruction Attacks and Countermeasures of DNN Training in Vertical Federated Learning](#)” , in *IEEE TDSC, Volume 22, Issue 3*
- 2024 Yongkang Zhang, Haoxuan Yu, Chenxia Han, Cheng Wang, Baotong Lu, Yunzhe Li, **Zhifeng Jiang**, Yang Li, Xiaowen Chu, Huaicheng Li. “[SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs](#)” , in the *Proc. of ACM PPOPP 2025*
- 2024 Na Lv, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Jiayi Zhang, Quan Chen, Minyi Guo. “[FedCA: Efficient Federated Learning with Client Autonomy](#)” , in the *Proc. of ICPP 2024*
- 2021 Minchen Yu, **Zhifeng Jiang**, Hok Chun Ng, Wei Wang, Ruichuan Chen, Bo Li. “[Gillis: Serving Large Neural Networks in Serverless Functions with Automatic Model Partitioning](#)” , in the *Proc. of IEEE ICDCS 2021; Best Paper Runner-Up*
- 2021 **Zhifeng Jiang**, Wei Wang, Yang Liu. “[FLASHE: Additively Symmetric Homomorphic Encryption for Cross-Silo Federated Learning](#)” , in *arXiv preprint (Citation: 108+)*

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#### Honors and Awards

- 2024, 2023 Redbird Academic Excellence Award, HKUST
- 2021 Best Paper Runner-Up Award (Top 3 out of 97 accepted papers), IEEE ICDCS
- 2019 Outstanding Graduate Award (Top 1%), Zhejiang Province
- 2017 He Zhijun Scholarship (Top 10 in Dept. of CS), ZJU
- 2017 National Scholarship (Top 0.1% nationwide), Ministry of Education, China

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#### Professional Service

- Journal Reviewer [IEEE TMC](#), [IEEE TBD](#), [IEEE TNNLS](#), [IEEE TKDE](#), [IEEE TDSC](#), [IEEE TC](#).
- Conference Reviewer [ICLR 2026 Workshop RSI](#), [ACM EuroSys 2023 \(Shadow\)](#).
- AEC Member [USENIX OSDI 2022](#), [USENIX ATC 2022](#), [ACM SOSP 2021](#).

### 研究兴趣

**具身智能系统**, 尤其关注视觉-语言-动作 (VLA) 模型、真实机器人学习、第一人称可穿戴智能体、机器人智能体, 以及长程操纵任务。

**端侧AI 系统**, 尤其关注面向手机的大语言模型 (LLM) 推理加速、面向AI 眼镜的语音增强, 以及面向个人电脑的工具调用智能体。

### 经历

- 2024 年11 月– 至今 **中央软件院研究员**,  
华为技术有限公司, 深圳, 中国
- 具身智能与机器人智能体
    - 带领6 人团队构建 **RobiAgent**, 一个双臂机器人智能体原型, 用于研究真实机器人操纵, 以及存在实时约束的耦合机器人子任务协调。
    - 围绕基于VLA 的机器人操纵, 完成 $\pi_{0.5}$  **仿真复现**、**真实机器人**数据采集、模仿学习, 以及在SO101 机械臂上的部署。
  - AI 眼镜上的可穿戴多模态智能体
    - 构建 **VisionBot**, 一个面向AI 眼镜的常驻感知与服务智能体, 已完成基于连续视觉搜索的物品寻找, 以及基于第一人称活动记录和反馈的健身指导两个demo。
    - 开发面向AI 眼镜的定向拾音系统, 包括设计双声道语音增强模型、构建流式推理流水线, 并通过HarmonyOS 应用完成部署。
  - 大语言/视觉语言模型应用与系统
    - 开发大语言/视觉语言模型应用, 包括端侧涉诈网页检测、大规模检索增强问答, 以及兼容MCP 的通用命令行智能体 (**Hey**)。
    - 探索通过并行解码加速自回归语言模型的推理。
- 2019 年9 月– 2024 年7 月 **计算机科学与工程博士研究生**,  
香港科技大学, 九龙, 中国香港
- 大语言模型推理的隐私保护
    - 在尽量降低对话质量损失的前提下保护系统提示词 (**PromptKeeper**)。
  - 隐私保护与高效联邦训练
    - 在存在恶意服务器的场景下, 保护联邦学习参与方选择过程的安全性 (**Lotto**)。
    - 在不损失时间性能的前提下, 实现可抵抗客户端掉线的差分隐私联邦学习 (**Dordis**)。
    - 基于异步训练原则实现端到端训练加速 (**Pisces**)。

### 教育背景

- 2019 年9 月– 2024 年7 月 **计算机科学与工程博士**,  
香港科技大学, 九龙, 中国香港  
博士论文: “[Towards Private and Efficient Cross-Device Federated Learning](#)”
- 2018 年7 月– 2018 年10 月 **Prof. Dean Tullsen 研究组科研实习生**,  
加州大学圣地亚哥分校, La Jolla, 美国
- 研究x86-64 平台上针对返回导向编程攻击的**防御方法**。
- 2015 年9 月– 2019 年6 月 **计算机科学与技术学士**,  
浙江大学, 杭州, 中国  
GPA: 3.97/4.0, 获浙江省优秀毕业生

### 论文发表

代表论文

- 2025 **Zhifeng Jiang**, Zihua Jin, Guoliang He. “[PromptKeeper: Safeguarding System Prompts for LLMs](#)” , *Findings of the ACL EMNLP 2025*
- 2024 **Zhifeng Jiang**, Peng Ye, Shiqi He, Wei Wang, Ruichuan Chen, Bo Li. “[Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning](#)” , *USENIX Security 2024*
- 2024 **Zhifeng Jiang**, Wei Wang, Ruichuan Chen. “[Dordis: Efficient Federated Learning with Dropout-Resilient Differential Privacy](#)” , *ACM EuroSys 2024*
- 2023 **Zhifeng Jiang**, Wei Wang, Bo Li, Qiang Yang. “[Towards Efficient Synchronous Federated Training: A Survey on System Optimization Strategies](#)” , *IEEE TBD* 第9 卷第2 期
- 2022 **Zhifeng Jiang**, Wei Wang, Baochun Li, Bo Li. “[Pisces: Efficient Federated Learning via Guided Asynchronous Training](#)” , *ACM SoCC 2022*

### 其他论文

- 2026 Na Lv, Jiayi Zhang, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Quan Chen, Minyi Guo. “[Enabling Client-Autonomous Training Optimizations for Efficient Federated Learning](#)” , *IEEE TPDS*
- 2026 Zekai Su, Yongjun He, Jiqing Han, Jianchen Li, Yikun Jiang, **Zhifeng Jiang**, Tianhong Ding. “[AS-EVNorm: Tail-Aware Extreme Value Normalization for Speaker Verification](#)” , *IEEE Signal Processing Letters* 第33 卷
- 2026 Jiayi Zhang, Zuo Gan, Chen Chen, **Zhifeng Jiang**, Hao Wang, Yifei Zhu, Quan Chen, Minyi Guo. “[Boosting Gradient-based Training Diagnosis for Efficient and Accurate Federated Learning](#)” , *IEEE TMC*
- 2024 Peng Ye, **Zhifeng Jiang**, Wei Wang, Bo Li, Baochun Li. “[Feature Reconstruction Attacks and Countermeasures of DNN Training in Vertical Federated Learning](#)” , *IEEE TDSC* 第22 卷第3 期
- 2024 Yongkang Zhang, Haoxuan Yu, Chenxia Han, Cheng Wang, Baotong Lu, Yunzhe Li, **Zhifeng Jiang**, Yang Li, Xiaowen Chu, Huaicheng Li. “[SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs](#)” , *ACM PPOPP 2025*
- 2024 Na Lv, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Jiayi Zhang, Quan Chen, Minyi Guo. “[FedCA: Efficient Federated Learning with Client Autonomy](#)” , *ICPP 2024*
- 2021 Minchen Yu, **Zhifeng Jiang**, Hok Chun Ng, Wei Wang, Ruichuan Chen, Bo Li. “[Gillis: Serving Large Neural Networks in Serverless Functions with Automatic Model Partitioning](#)” , *IEEE ICDCS 2021*; 获最佳论文奖提名
- 2021 **Zhifeng Jiang**, Wei Wang, Yang Liu. “[FLASHE: Additively Symmetric Homomorphic Encryption for Cross-Silo Federated Learning](#)” , *arXiv* 预印本 (引用108+)

### 荣誉奖励

- 2024, 2023 香港科技大学红鸟学术卓越奖
- 2021 IEEE ICDCS 最佳论文提名奖 (97 篇录用论文中前3)
- 2019 浙江省优秀毕业生 (前1%) , 浙江省教育厅
- 2017 浙江大学何志均教育基金奖学金 (计算机学院前10)
- 2017 国家奖学金 (全国前0.1%) , 中华人民共和国教育部

### 专业服务

- 期刊审稿人 [IEEE TMC](#), [IEEE TBD](#), [IEEE TNNLS](#), [IEEE TKDE](#), [IEEE TDSC](#), [IEEE TC](#).
- 会议审稿人 [ICLR 2026 Workshop RSI](#), [ACM EuroSys 2023 \(Shadow\)](#).
- Artifact 评审委员会成员 [USENIX OSDI 2022](#), [USENIX ATC 2022](#), [ACM SOSP 2021](#).